

HemisFear: A Virtual Reality Exposure Therapy Prototype Driven by Cross-Cultural Understanding of Dog Phobia

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Figure 1: A low-intensity example of virtual exposure to Indian stray dogs in *HemisFear*.

Abstract

Cynophobia, the fear of dogs, is a common and debilitating phobia. Live exposure therapy is first-line treatment for many phobias, but using live dogs may be expensive and unpredictable. Virtual reality exposure therapy (VRET) is a promising alternative, but remains underexplored for cynophobia. Furthermore, existing VRET designs seldom account for cultural backgrounds, which influence cynophobia's fear factors and severity. For example, stray dog encounters in South Asia pose unique risks compared to domestic dogs in the Anglosphere. This paper introduces *HemisFear*, a VRET prototype featuring two culturally informed environments: an Anglo-Western residence with domestic dogs, and an Indian city with strays. The Western environment was shaped by a content analysis of prior VRET literature, while participatory design interviews with individuals from India ($N = 4$) informed the stray dog environment.

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By reconciling VRET principles with cultural sensitivities, *HemisFear* may offer a path towards effective and inclusive treatment for cynophobia.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Participatory design**; **Scenario-based design**; • **Applied computing** → **Psychology**.

Keywords

Virtual reality, human-dog interaction, exposure therapy, cynophobia

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1 Introduction

Cynophobia, the fear of dogs, is one of the most common animal phobias worldwide, as 36% of animal phobia patients report fearing either dogs or cats [16]. Cynophobia is often more debilitating than other phobias, as dogs are ubiquitous in many countries. For example, 33% of Canadian households had a pet dog in the year 2009, and

there may be as many as 70 million stray dogs in India [7, 15]. It can be difficult for cynophobic individuals to avoid encountering dogs in daily activities; thus, accessibility and availability of treatment is crucial. A scoping review of phobia treatment notes that exposure therapy — gradual interaction with anxiogenic stimuli — is the first line of evidence-based treatment for many phobias [12]. Exposure therapy for cynophobia involves interacting with dogs in controlled settings; however, the use of real dogs presents feasibility and safety concerns. For example, dogs are sensitive to certain human emotions, including fear, and may adjust their behaviour accordingly [4]. This suggests that even a well-trained dog's behaviour may not be fully controllable; in the event of an adverse reaction, cynophobic individuals could become re-traumatized. Furthermore, training and day-to-day care of the dogs is cost-prohibitive and time-consuming [16].

In situations where in-person exposure is infeasible, virtual reality exposure therapy (VRET) is often an effective alternative. Virtual reality (VR) can replicate real-world exposure while eliminating physical risks, allowing users to experience stimuli without danger [11]. Additionally, VRET software can be programmed to delegate control over anxiety-inducing stimuli to either the clinician or VR user [11]. This enables therapy to be *tailored*: the intensity of the stimulus can be adjusted to match a given individual's needs at any stage of their therapy journey. Due in part to these advantages, VRET typically displays similar efficacy to in-person exposure in comparative studies [1].

However, VRET is largely unexplored in the context of cynophobia. A prior systematic scoping review of virtual human-dog interaction by Oxley et al. found just six peer-reviewed articles pertaining to VR exposure therapy for cynophobia [13]. This results in the absence of evidence-based design principles for human-dog interactions in VR, especially with respect to exposure therapy. In addition, fear factors and intensity of cynophobia can significantly differ across cultural backgrounds [5]. For example, in Euro-Western countries, cynophobia may be characterized by misinterpretation of domestic dog body language, or incorrect assumptions that dog aggression is unprovoked [13]. Conversely, cynophobia in India and South Asia (hereafter called Indo-Asia) is influenced by the aforementioned prevalence of unsocialized stray dogs, as well as the threat of rabies transmission [7]. Since five of the six prior studies were conducted in either Western Europe or Australia, the current VRET design space does not adequately accommodate experiences of cynophobia beyond the Western hemisphere [13]. This monocultural lens may no longer be suitable even for Euro-Western countries due to modern globalization; the population of Indian immigrants in Canada has more than quadrupled since 2013, for example [3].

We thus present a VRET prototype titled *HemisFear*, which integrates fear factors of cynophobia from both Euro-Western and Indo-Asian bases. Two virtual environments (VEs) were built for the *HemisFear* prototype: the first depicts an Anglo-Canadian residence and affords interaction with domestic dog models, while the other resembles an Indian city and affords interaction with stray dogs. Design principles for the domestic dog environment, including variance in the dog's breed, behaviour, and proximity to the player, were established following a content analysis of prior cynophobia-focused VRET studies. Key components of the stray

dog environment, including varying numbers of stray dogs and real-time behavioural instructions for the user, were extracted from semi-structured ethnographic interviews with individuals from India ($N = 4$). By addressing the two aforementioned gaps in cynophobia-based VRET literature, *HemisFear* illuminates a potential path towards accessible, inclusive, and effective treatment for the fear of dogs.

2 Related Literature

2.1 Indo-Asian Attitudes Towards Dogs

To the best of the authors' knowledge, this paper is the first cross-cultural analysis of cynophobia as it specifically pertains to exposure therapy techniques. However, the general effects of cultural background on attitudes towards dogs has been explored in several prior studies, and evidence suggests that individuals from South Asia may view dogs more negatively than individuals in the Western hemisphere. For example, certain religious customs in Indo-Asian countries depict dogs as unclean or untouchable "pariahs" of society [18]. Individuals from such countries who practice these customs may be more likely to avoid dogs than North American individuals, whose culture views the domestic dog as the proverbial "man's best friend" [18].

The above sentiment was also suggested by a previous comparative study on experiences of animal phobias in seven Western and Asian countries [5]. Among the studied phobias was cynophobia, and both the United States and India were represented by the study's participants. Univariate tests for the fear of dogs between countries revealed that Indian participants were, on average, significantly more fearful of dogs than Americans [5]. The religious perspectives described in [18] may contribute to this fear, as may the prevalence of both stray dogs and rabies. Nearly 70% of all human rabies cases occur in India, which, combined with rabies vaccine accessibility issues in small towns and villages, contributes to Indian civilians' fear of free-roaming dogs [7].

Furthermore, a prior study on dog behaviour recognition suggests that the ability for humans to understand dogs' emotional cues strengthens with experience [23]. The researchers observed that happy dog behaviours could typically be identified regardless of prior experience, but experienced participants were significantly better at identifying fearful dog behaviours than inexperienced participants. This may be an additional factor of Indo-Asian cynophobia, as both stray and guard dogs are more common than domestic dogs in India [9]. Thus, some individuals from India may lack sufficient basis with domestic dogs to accurately understand their behaviour.

2.2 VRET for Cynophobia

A search of the PubMed, Institute of Electrical and Electronics Engineers (IEEE), and Association for Computing Machinery (ACM) databases identified six articles on VRET prototypes for cynophobia. Exclusion criteria included non-English articles and studies that did not use an immersive head-mounted display (HMD), as its absence suggests a lack of three-dimensionality. Oxley et al. note that immersion and navigation ability are often weaker in 2D VR scenarios compared to 3D environments [13].

Of the six articles, only two assessed therapeutic merit, while the remaining four examined affective responses to virtual dog stimuli.

2.2.1 Therapeutic Merit Studies. Farrell et al. (hereafter called **F**) created a VRET prototype based on 360-degree footage of six breeds. The dogs performed a range of behaviours, including sitting, walking, and running both on and off leash [6, 13]. Children ($N = 8$) with a clinical diagnosis of dog phobia underwent four weekly sessions of exposure therapy in F's prototype; during their one-month followups post-treatment, seven of eight reported a newfound ability to approach and pet real dogs [6].

Alves and Esteves (hereafter called **AE**) explored the potential benefits of *gamification* – the incorporation of game-resemblant elements into a task to enhance engagement and motivation – and real-time heart rate feedback in both 360-degree video and full VRET for cynophobia. Participants ($N = 10$) experienced comparatively higher anxiety in the full VR environment, improved self-control and lower heart rates when biofeedback was present, and higher engagement with gamified elements [2].

2.2.2 Affective Response Studies. Viaud-Delmon et al. (hereafter called **VD**) were the first to publish a VRET-oriented exploration of cynophobia fear factors. VD assessed the tendency for various spatialized auditory stimuli, such as dog sounds, to evoke anxiogenic responses. Upon evaluating three VEs with differing levels of audio fidelity, they found that audio of dogs barking and growling elicited the highest levels of anxiety [22].

Taffou et al. (hereafter called **T**) compared the effects of *unimodal* (either auditory or visual) and *bimodal* (both auditory and visual) dog stimuli, as well as dog motion, on affective responses of VR users. In two behaviour assessment tests, participants consistently reported higher anxiety levels in response to bimodal stimuli than unimodal stimuli [21].

Suied et al. (hereafter called **S**) explored four potential fear factors: dog breed, motion, and physical proximity to the VR player; as well as multimodality. Their results were similar to those of T, indicating that dogs simulated with both audio and visuals were more anxiety-provoking than purely visual models [20].

Most recently, Hnoohom and Nateerataiwa (hereafter called **HN**) assessed the tendency for multimodality, dog model proximity, and dog model behaviour to provoke anxious responses in cynophobic participants. The third and final dog encountered in their prototype, which could jump and attack the user, received the highest average rating of “dreadfulness” [8].

VD, T, and S were conducted in France, while F occurred in Australia, AE in Portugal, and HN in Thailand. None of these studies disclosed additional participant demographic information.

3 Design Methodology

The development process for *HemisFear* was influenced by prior research regarding both general VRET efficacy and cross-cultural attitudes towards dogs. Of the six studies discussed in the previous section, none occurred in Indo-Asian countries, and merely one (HN) is situated outside the Western hemisphere. Thus, design requirements for the stray dog environment in *HemisFear* were predominantly established through participatory design interviews with individuals from India. Insights for the domestic

dog environment were extracted via a content analysis of existing cynophobia-focused VRET literature.

3.1 Content Analysis

To inform our creation process for *HemisFear*, we performed a five-factor content analysis on prior cynophobia VRET literature. The factors explored in this analysis included dog breeds used, tailorable interactions, VE setting, pacing and progress criteria, and fear assessment instruments used in the study; their overall focuses as detailed above are also included. Due to the dominant Euro-Western basis for most prior studies, insights generated from this analysis were only applied to *HemisFear*'s domestic dog environment. A tabular summary of the findings for each factor, which are otherwise discussed below, can be found in Appendix A, Figure 4.

3.1.1 Dog Breeds. There appears to be no consensus on what breeds are best suited for VRET. AE and F used common domestic breeds, such as Labrador mixes, while S and HN used more protective breeds such as Dobermans and Rottweilers. Moreover, while breeds are sometimes tailorable, there is no established basis for evaluating them comparatively. Oxley et al. note that due to this lack of “systematic reuse,” it is difficult to make meaningful comparisons between the psychometric effects of dog breed models [13]. They call for dog models to be selected, justified, and conveyed in detail within future studies.

3.1.2 Tailorability. Commonly tailorable elements include proximity of the virtual dog to the user, movement and behaviour of the virtual dog, spatial audio for both environmental ambience and dog sounds, and the range of tasks that the user must perform. In gamification-focused studies such as AE and VD, tasks include navigating to a location in the environment, retrieving an object, or moving along a certain route. The presence of virtual dogs as obstructions to task completion is also tailorable in such studies.

3.1.3 Environmental Setting. In four of six cases, multiple VEs were implemented. These environments typically reflect urban settings wherein domestic dogs may be encountered, such as yards, greeneries, or rooms of a house. The two exceptions to this design choice were F, who filmed 360-degree videos of different dogs in an indoor space, and S, who implemented one outdoor VE and tailored the intensity of fog in its sky. Thus, each study involved tailoring the virtual setting in some capacity as well.

3.1.4 Pace. The pace of a given exposure therapy session is usually determined by participant progress. F progressed tasks based on Subjective Unit of Distress (SUD) reductions, moving to the next intensity level when a participant's distress reduced by 50% with their current level [24]. AE, T, and VD only progressed tasks when a task was completed; as mentioned previously, participants responded favourably to this gamification in AE. Pacing in S was purely time-based; after 10 minutes of exposure to a given intensity level, participants would move to the next level. No information was reported by HN.

3.1.5 Instruments. SUD was used in five of the six studies, indicating that it should be considered standard practice for recording general affective responses to VRET stimuli. State-Trait Anxiety

Inventory (STAI) was used in AE, S, and VD to measure transient and persistent levels of anxiety [19].

3.2 Participatory Design Interviews

To extract design ideas for the stray dog environment, we performed semi-structured interviews with four invited participants. This study was approved by the institutional research ethics board, and participants provided their written informed consent. Three participants (P1, P2, P4) are graduate-level university students who spent either their formative years or early adulthood in India, but now live in Canada. The other participant (P3) moved from India to the United States after completing university. Additionally, P4 was interviewed post-submission to address reviewer feedback, which requested a larger sample and richer dataset.

Interviews were conducted via WhatsApp text messaging, lasted 15 minutes in length, and were guided by the following topics: prevalence of stray dogs in their area, attitudes towards stray dogs, negative experiences with stray dogs in their past, impact of stray dog experiences on behaviour towards Canadian domestic dogs, effectiveness of domestic dog-themed VRET, and stray dog-themed VRET design ideas.

3.2.1 Prevalence of Stray Dogs. P1 stated that stray dogs were “almost everywhere” in their neighbourhood. Stray dogs were common on P2’s campus and occasionally followed people who were alone on nearby streets. P3 also stated that dogs were common in their region in India and that stray dogs were viewed as “part of the environment rather than integrated family members.” Similarly, P4 described stray dogs as “pretty common” in their home region.

3.2.2 Attitude Towards Stray Dogs. P1 was afraid of stray dogs during childhood and would avoid streets with dogs, but their fear began to wane as they had positive experiences with friendlier strays. P1’s parents did not want dogs in or around their family home. P2 noted that although the dogs on their campus were typically friendly, packs of strays could be dangerous at night. P2 and their friends were instructed to “pretend they weren’t there” when walking past them, and P2’s grandparents also warned them to stay away from stray dogs due to rabies. P3 stated that the stray dogs in their childhood neighbourhood were sometimes unfriendly and territorial. As P3 reached adulthood, they developed some sympathy but still remained cautious around stray dogs, especially when they were in packs. P4 reported a mixed attitude toward stray dogs, expressing caution in India but also familiarity with certain strays who were regularly fed by their family. Over time, P4 became less anxious about stray dogs, though some of their Indo-Asian friends who had negative encounters remain fearful of dogs in general.

3.2.3 Negative Experiences with Stray Dogs. P1 reported being attacked by dogs as a child and stated that they knew other people who had been bitten. P2 recounted a story of a friend who was bitten while trying to feed a stray dog. P3 was chased and barked at aggressively while riding their bicycle, and knew of a neighbour whose child was bitten by a stray dog while playing outside. P4 also experienced being chased by a stray dog, but was able to escape before being bitten. They noted that their primary fear stemmed from the risk of requiring post-bite vaccination, rather than the dogs themselves.

3.2.4 Impact on Behaviour Towards Domestic Dogs in North America. P1 does not fear domestic dogs and attributes this to one of their relatives having a “really friendly dog.” P2’s lack of experience with domestic animals causes them to “err on the side of caution.” This individual does not interact with dogs unless they are approached by one who appears friendly. P3 was initially uneasy around large breeds and unfamiliar dogs due to perceived unpredictability, but soon realized that dogs in America are usually well-trained and socialized, and are thus “much more predictable.” P4’s experiences are similar to those of P3, although they remain uncomfortable around dogs who bark frequently.

3.2.5 Domestic Dog VRET Effectiveness. P1 believes that any individual who has had a traumatic experience with a dog could benefit from any kind of dog-related exposure therapy, as well as individuals with no prior experience of pet dogs. However, they believe that to treat the average Indian immigrant, the VRET should feature stray dogs. P2 hypothesized that VRET could help people like themselves, who do not fear dogs but also do not love them, and individuals with “legitimate phobia.” P3 explained that incorporating stray dogs and other “Indian contexts” may be necessary in order to address triggers related to unpredictable and aggressive behaviour. P4 agreed that VR-based exposure therapy could “help regulate the immediate FFF (fight, flight, freeze) response,” but suggested that experiences with real dogs would ultimately be necessary in order to establish trust.

3.2.6 VRET Design Ideas. P1 suggested that stray dogs should be a component of the VRET software. They recommended that in at least one condition, the dogs should follow the player as they walk away, and that a pack of dogs would be most realistic if the player is on a vehicle (such as a bicycle). P2 noted that a pack of stray dogs as a high-intensity level would be realistic to their region. They recommended that the VRET application train users when to engage and when to ignore stray dogs, as some are rabid. An interaction wherein a stray dog walks up to the player and the player must choose from three responses – doing nothing, petting it, or walking away – was proposed. P3 suggested that the therapy begin with stray dogs so that past traumas could be addressed, and gradually transition to domestic dogs to help Indian immigrants “adapt to their new surroundings.” Each participant responded favourably to the idea of basing the environment on a populous Indian city. P4 supported the use of stray dogs for exposure therapy and “reliving [traumatic] experiences,” but also argued that domestic dogs of various dispositions may be better for real-world “training.”

3.3 Cross-Cultural Design Considerations

The insights from our participatory design interviews revealed several key considerations for designing culturally sensitive VRET. These considerations, which influenced our design of *HemisFear*, are presented as actionable design guidelines below:

- (1) **Culture-specific dogs may not always be necessary.** Participants agreed that exposure to virtual domestic dogs could be beneficial regardless of cultural background. However, multiple participants emphasized the distinction between trauma-focused and training-focused exposure. For individuals whose fear stems from traumatic interactions with stray

dogs, exposure to domestic dogs alone – especially in the context of training for future interactions – may be insufficient.

- (2) **When culture-specific dogs are used, the virtual environment should reflect the same context.** Several participants supported the idea of situating stray dogs in an urban Indian setting, as this more accurately reflects scenarios wherein stray dogs are encountered. Multiple participants also suggested that VR exposure to pet dogs in domestic environments would be useful in later stages of therapy, especially for users who are adapting to a new region and cultural context.
- (3) **Dog behaviour should be region-accurate.** Participants described stray dogs in South Asia as territorial, prone to moving in packs, and likely to chase individuals, particularly those on bicycles. Some also noted that strays tend to follow people who are alone or attempt to ignore them. These behaviours differ from those of North American domestic dogs, and may reflect past traumas experienced by Indo-Asian individuals with cynophobia. This suggests that in order to ensure ecological validity, virtual dogs should exhibit behaviours that reflect realistic encounters in their native regions.

4 HemisFear Implementation Details

4.1 Anglo-Western Domestic Dog Environment

In thematic alignment with prior studies, the chosen setting is the balcony, yard, and exterior surroundings of a North American house. The user must complete a series of tasks which often involve observing, passing, or interacting with dogs in some capacity. These tasks escalate in intensity, and each task must be completed in order to advance to the ensuing one. Tailored difficulty is afforded along the following four dimensions: breed, proximity, behaviour, and tasks.

- (1) **Breed:** The Labrador Retriever, German Shepherd, and Pit Bull breeds were selected based on recognizability as well as availability of high-quality 3D models in Unity. Models from prior VRET studies were not re-used in *HemisFear* due to a lack of information regarding their availability. In response to a gap identified by Oxley et al., information regarding the dog models used in *HemisFear* can be found in Appendix B [13].
- (2) **Proximity:** Direct interaction between user and dog is impossible at the lowest intensity level, as the user is positioned on the balcony while the dog is in the yard (Figure 2, left). At moderate to high intensities, proximity is indirectly controlled by the dog’s behavioural parameters as explained below.
- (3) **Behaviour:** At lower intensity levels, the dogs are often stationary, which may involve sleeping (see Figure 2, centre), lying down, or standing still. As intensity increases, the dogs begin to move more regularly and with faster speed, often showing increased interest in the user. They do not behave aggressively, but occasionally attempt to “play,” such as in Figure 2 (right).

- (4) **Tasks:** Initial tasks involve simply observing the dog for a predetermined duration, but eventually escalate to *traversal* – moving to a certain location – and *search* – finding a signified object and returning it to its proper location, as seen in Figure 2 (right). Each task must be completed in order to progress through the entire therapy module.



Figure 2: Three different intensity settings for the domestic dog environment. Left: A low-intensity example wherein the player views a Labrador Retriever from the balcony. **Centre:** A medium-intensity example wherein the player stands in front of a sleeping Pit Bull Terrier. **Right:** A high-intensity example in which the player must retrieve an object (indicated by a pink halo) while a Labrador Retriever attempts to play with them.

4.2 Indo-Asian Stray Dog Environment

In accordance with the design guidance provided by participants, the chosen setting is an “Idyllic Indian City” environment (see Appendix B). Multiple models of stray dogs were used, with all three appearing as a pack in the scenario’s highest intensity level. The structure and tailorable interactions are similar to those of the domestic dog environment, with stray dogs being stationary and uninterested in the player at lower intensity levels (see Figure 3, left), and having a higher tendency to move or show interest in the user at higher levels. The gamified tasks combine search or traversal with instructions on how to behave around or towards the stray dogs. For example, in Figure 3 (right), the player is tasked with walking to a signified storefront and acting as if the stray dogs are not present; other actions include standing still or slowly backing away.



Figure 3: The stray dog environment in *HemisFear*. Left: A low-intensity setting involving observation of a stray dog from afar. **Right:** A high-intensity setting involving a search task (indicated by the pink halo) and a pack of stray dogs.

5 Outlook and Future Work

While the *HemisFear* prototype reflects a culturally informed approach to VRET for cynophobia, several limitations should be discussed. For example, the participatory design interviews were conducted on a small sample size ($N = 4$), which may limit the generalizability of our findings. As well, no participant reported a clinical diagnosis of cynophobia, so their insights may not reflect the needs of those seeking treatment. Future explorations of *HemisFear* should incorporate a larger pool of participants with different intensities of cynophobia.

Next, although WhatsApp is accessible and convenient, its messages cannot capture nonverbal details, such as facial expressions or intonation. Future research would benefit from a transition to in-person or video-based interviews, which could generate more nuanced insights regarding user needs and preferences.

Furthermore, the current *HemisFear* prototype uses 3D models with inconsistent quality and artistic style. If these disparities are noticed by users, their affective responses – including anxiety and distress – may be influenced. Future work should investigate this potentially confounding effect. If necessary, subsequent iterations could then be designed with a uniform artistic style and strict model quality requirements.

Nonetheless, the creational structure of *HemisFear* – as well as its conceptual adherence to multicultural insights on cynophobia – does lend itself to a randomized controlled study that evaluates two key efficacy parameters. In alignment with prior studies, *HemisFear*'s therapeutic effectiveness could be measured via the Subjective Unit of Distress (SUD) or State-Trait Anxiety Inventory (STAI), or both [19, 24]. These instruments would be administered pre- and post-exposure across each VRET session, and again during follow-up evaluation some time after the patient's final session.

Next, certain usability and feasibility characteristics of the VRET software are typically evaluated alongside efficacy. For example, the ability for users to experience the sense of *presence* ("being there") in the virtual environment is a key contributor to positive VRET outcomes, as it often indicates realism and ecological validity [14]. In prior cynophobia-focused VRET studies, presence has been measured with the Igroup Presence Questionnaire (IPQ) [17]. High IPQ scores for *HemisFear* may thus be predictive of therapeutic efficacy. A second component of VRET usability concerns *VR sickness*, a condition similar to motion sickness that is triggered by the illusion of self-motion [10]. VR sickness in *HemisFear* may result from controller-based locomotion or scene transitions, so future work could incorporate Virtual Reality Sickness Questionnaire (VRSQ) measurements [10]. If high VRSQ scores are observed, then mitigation techniques may need to be implemented in *HemisFear*, as VR sickness is associated with study withdrawals.

Beyond feasibility and pilot efficacy, future work could investigate how *HemisFear*'s design methodology might generalize to larger and more diverse participant samples. This could include different cultures beyond the Anglosphere, Western Europe, and South Asia. Moreover, different dog breeds and behaviours, participant tasks, and virtual environments could be integrated and evaluated.

In this paper, we introduced *HemisFear*, a VRET prototype for cynophobia that addresses both the scarcity of related literature

and the absence of cultural considerations in prior solutions. By accommodating insights from both Euro-Western and Indo-Asian perspectives, *HemisFear* contributes to a more accessible and inclusive design space for treating cynophobia – a common and often debilitating condition. Further research should evaluate *HemisFear*'s usability, feasibility, and efficacy in Euro-Western and Indo-Asian populations, and eventually assess its adaptability to other cultural contexts and cynophobic fear factors as well.

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Appendix

A Content Analysis Summary

Study	Breeds	Tailorability	Setting	Pace	Instruments
AE [2]	Not mentioned; Golden Retriever, Black Labrador Retriever indicated by screenshots	Visual fidelity (360-degree vs. 3D), tasks (retrieve object, move, pet the dog), biofeedback	4 total: indoor office, living room, fenced yard, open field	Task-driven, but breaks were taken between conditions	<i>Subjective Unit of Distress</i> (SUD), <i>State-Trait Anxiety Inventory</i> (STAI)
F [6]	Doberman, Cocker Spaniel, Labrador-Kelpie, Border Collie-Rottweiler	Restraint (leash/no leash), dog motion, proximity	360-degree footage of university therapy room	SUD taken every 5 minutes; task advanced upon 50% SUD reduction	SUD
HN [8]	Rottweiler	Proximity (in kennel/free), audio, behaviours (run, jump, growl, attack)	3 total: house/garden areas	Not mentioned	4-point Likert scale of 'dreadfulness'
S [20]	Doberman, Malamute, Mini Pinscher	Environmental fog intensity; audio; dog motion, colour, and proximity	Outdoor scene (dallight, fog, ambient)	10 minutes per session	14-question dog phobia survey; SUD, STAI
T [21]	Not mentioned	Sensory modality (unimodal vs. bimodal), dog motion, audio	3 total: corridor, garden, dark hangar	Task-focused: find a frog in each environment	SUD
VD [22]	Doberman (selected by participants from nine total options)	Tasks (find objects/locations), dog behaviour (sit/stand), audio	3 total: city, garden, dark hangar	Task-focused: find object/location	SUD, STAI

Figure 4: Content Analysis of Existing Cynophobia-Based VR Literature

B Design Resources

- German Shepherd Unity asset
- Idyllic Indian City Unity asset
- Labrador Retriever Unity asset
- Pit Bull Terrier Unity asset
- Stray Dog models